

P06: Classification and Characterisation of Encrypted IoT Traffic Using Network-Level Features

Proponents: Paulo Carvalho (pmc at di.uminho.pt), Solange Rito Lima (solange at di.uminho.pt), Pedro Sousa (pns at di.uminho.pt), Flávio Silva (fos at di.uminho.pt)

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Introduction

Internet of Things (IoT) deployments increasingly rely on encryption mechanisms such as TLS and DTLS to ensure data confidentiality and integrity. While encryption is essential from a security and privacy standpoint, it significantly reduces payload visibility for network operators, complicating traffic analysis, network management, and security monitoring.

Despite payload encryption, IoT traffic still exposes observable information at the network and transport layers, including packet sizes, timing patterns, and flow-level statistics. These features may reveal meaningful information about device roles, communication patterns, and application behaviour, even when content remains inaccessible.

This project investigates the extent to which encrypted IoT traffic can be classified and characterised using network-level features only, without decrypting payloads. The focus is on understanding traffic patterns generated by different IoT devices and services, and on assessing what information remains observable under encryption, as well as the associated limitations and privacy implications.

Objectives

The main objective of this project is to analyse and classify encrypted IoT traffic based on flow-level and network-level features, and to evaluate the feasibility and limitations of traffic characterisation without payload inspection.

Specific objectives include:

- identify representative IoT traffic classes (e.g. periodic telemetry, event-driven messages, firmware updates)
- generate and capture encrypted IoT traffic in a controlled environment
- extract and analyse relevant network- and flow-level features
- evaluate the separability of IoT traffic classes under encryption
- analyse robustness under varying network conditions (e.g. delay, loss, rate limitation)
- discuss implications for network management, monitoring, and user privacy

Payload inspection is neither required nor expected; the focus is strictly on observable network-level features.

Students may frame the work around questions such as:

- which network-level features are most informative for characterising encrypted IoT traffic?
- to what extent can different IoT traffic classes be distinguished under encryption?
- how robust are classification results under changing network conditions?
- what are the practical limits of traffic visibility in encrypted IoT environments?

Expected Results

The expected results include:

- a structured characterisation of encrypted IoT traffic types
- identification of relevant network- and flow-level features for traffic classification
- quantitative evaluation of classification performance and limitations
- analysis of robustness under varying network conditions
- a final technical report documenting methodology, experiments, results, and conclusions

Planning

Suggested timeline:

Tarefa	Semana (2º Semestre)															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
T0: topics presentation and selection																
T1.1: literature review																
T1.2: traffic generation and capture																
T2: features extraction and exploratory analysis																
T3: traffic classification and characterisation																
T4: evaluation and robustness analysis																
T5: Project documentation, tech report																
Final assessment																

Obs. Suggestions

Students may use any suitable open-source tools. Typical examples include:

- IoT protocols: MQTT over TLS, CoAP over DTLS, HTTPS-based APIs
- traffic sources: IoT laboratory devices, simulators, or controlled generators
- feature extraction: flow statistics, timing features, packet size distributions
- tools: tcpdump, tshark, Zeek (optional)
- data analysis: Python libraries (pandas, NumPy, matplotlib, seaborn, SciPy, scikit-learn) for data processing, visualisation, and classification.
- classification approaches: statistical methods or lightweight machine learning (e.g. NFStream, Cisco Joy, nPrint, or equivalent frameworks). The use of ML techniques is optional, i.e., they may be used as an exploratory tool. If used, the focus should be on feature relevance, interpretability, and robustness, rather than maximising classification accuracy.